

ABHISHEK GOND

Software Engineer | AI & GenAI Engineer | Full-Stack Developer

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EDUCATION

• Institute of Engineering and Technology

B.Tech in Electronics and Communication Engineering; CGPA: 7.85/10

Lucknow, India

2023 – 2027 (Expected)

EXPERIENCE

• Software Engineer Fellow (Fellowship)

Jul 2026

Panscience Innovations | Node.js, Express 5, TypeScript, Next.js, MongoDB, Pinecone, RAG, LLM

- Built an Express 5/TypeScript backend for multimodal clinical PDF extraction, verifying each value against source text and handling LLM failures via fallback and circuit breaker.
- Designed a Pinecone RAG system with local BGE-small embeddings, JWT tenant isolation, and MedlinePlus/MeSH, ACC/AHA and WHO guideline knowledge bases.
- Enforced citation validation and `insufficient_evidence` handling in code to block hallucinated references; validated with 270 backend and 10 frontend tests.

• Full-Stack Developer Intern

Feb 2026 – Apr 2026

Durga Foundation | Next.js, Node.js, Prisma, Redis, MySQL

- Migrated the NGO portal to Next.js and built the “Youth Corner” full-stack platform (Node.js, Express, MySQL).
- Architected an 11-client multi-schema Prisma setup; integrated Redis caching and Razorpay donation processing.

PROJECTS

• GridSense – AI IoT Telemetry Gateway | Next.js, Node.js, PostgreSQL, Socket.IO, Prisma, pgvector Live Demo

- Built a **Socket.IO** pipeline streaming simulated panel readings every **2 s** to per-user rooms, batching 30 readings per panel into one Prisma write per minute.
- Designed a **LangGraph tool-calling loop** where Groq Llama 3.1 8B selects Zod-validated Prisma tools instead of generating SQL, plus pgvector retrieval over 384-d MiniLM embeddings.
- Implemented rotating **httpOnly refresh-token sessions** (15-min JWTs) and Razorpay HMAC-SHA256 verification with an atomic Prisma transaction across payment, subscription and user records.

• Algorithmic RAG Engine (Framework-Free) | Python, NumPy, FastAPI, Streamlit, Gemini Live Demo

- Built a framework-free RAG pipeline in Python: O(N) sliding-window chunker (200-word windows, 50-word overlap) feeding local CPU MiniLM embeddings, with no LangChain or LlamaIndex.
- Implemented exact top-k retrieval via vectorized NumPy cosine similarity over an in-memory (M×384) float32 matrix, with the model pre-warmed at FastAPI startup.
- Designed FastAPI ingest/query endpoints separating retrieval from generation; Gemini answers under a strict grounding prompt with model fallback and per-stage latency reporting.

• Meeting & Video Intelligence Pipeline | Python, Streamlit, LangChain, Whisper, Mistral, ChromaDB Live Demo

- Developed an audio pipeline with yt-dlp and ffmpeg (16 kHz mono, 10-min chunks), routing by language between local Whisper and Sarvam AI (25s pieces for API limits).
- Implemented Mistral 8B map-reduce summarization via 3 LCEL chains (actions, decisions) on chunked transcripts (3000 chars, 200 overlap), paired with a MiniLM/ChromaDB RAG chat layer (top-4) and hallucination fallbacks.

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, SQL
- **Frontend:** HTML/CSS, React.js, Next.js, Tailwind CSS, Streamlit
- **Backend:** Node.js, Express.js, FastAPI, REST APIs
- **AI/ML & GenAI:** Machine Learning, LLMs, GenAI, RAG, LangChain, LangGraph, AI Agents, Vector Search
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Prisma, pgvector, ChromaDB, Pinecone
- **Tools & Core CS:** Git, Docker, CI/CD, OOP, DBMS, Operating Systems, Computer Networks

HACKATHONS & COMPETITIONS

• Hack Devengers 1.0 – Top 30 / 4.2K+ Participants – (CodeRAG Vector)

Live Demo | Aug 2026

- Engineered a FastAPI repository-audit pipeline ingesting local paths or GitHub zipballs, with directory pruning, extension whitelisting and 15-file/300-line caps to bound LLM token usage.
- Implemented RAG grounding with a 17-rule CS/security knowledge base (all-MiniLM-L6-v2) in ChromaDB, retrieving top-5 rules per source file.
- Integrated Gemini JSON-mode output validated by Pydantic with heuristic fallbacks and pytest test coverage.

• Smart India Hackathon (SIH) Internal Round – 3rd Rank

Sep 2026

- Architected **AGNI.AI**, a disaster response system fusing NASA FIRMS satellite telemetry, PostGIS mapping, and multimodal AI (VLM + XGBoost) to classify industrial thermal anomalies and trigger alerts.

ACHIEVEMENTS

- Data Structures & Algorithms: Solved 450+ problems on LeetCode and GFG over 100+ active days, focusing on time/space complexity optimization.

VOLUNTEER EXPERIENCE

• Core Member, Training & Placement Cell (TPC)

Sep 2025 – Present

Student liaison between hiring teams and candidates; guided peers through the campus recruitment process.